Reading seminar 3: "Stable group theory and approximate subgroups", by Ehud Hrushovski

Terence Tao · 29 October 2009

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This week, Henry Towsner continued some model-theoretic preliminaries for the reading seminar of the Hrushovski paper, particularly regarding the behaviour of wide types, leading up to the main model-theoretic theorem (Theorem 3.4 of Hrushovski) which in turn implies the various combinatorial applications (such as Corollary 1.2 of Hrushovski). Henry's notes can be found here.

A key theme here is the phenomenon that any pair of large sets contained inside a definable set of finite measure (such as $X \cdot X^{-1}$) must intersect if they are sufficiently "generic"; the notion of a wide type is designed, in part, to capture this notion of genericity.

— 1. Global types —

Throughout this post, we begin with a countable structure M of a language L, and then consider a universal elementary extension $\mathbb{U}$ of M (i.e. one that obeys the saturation and homogeneity properties as discussed in Notes 1. Later on, L will contain the language of groups, and then we will rename $\mathbb{U}$ as G to emphasise this.

Recall from Notes 1 that a *partial type* over a set A is a set of formulae (with n variables for some fixed n) using A as constant symbols, which is consistent and contains the theory of M; if the set of formulae is maximal (i.e. complete), then it is a *type*. One can also think of a type as an ultrafilter over the A-definable sets; if p is a type and B is an A-definable set given by some formula ϕ, then either ϕ lies in p (in which case we write $p \subset B$) or $\neg\phi$ lies in p (in which case we write $p \subset \overline{B}$) but not both.

When the set A is small (i.e. has cardinality less than that of $\mathbb{U}$, which in particular would be true of A consisted of M union with a finite set, which is a very typical situation), then by saturation one can identify types (or partial types) with the subset $p(\mathbb{U})$ of $\mathbb{U}^n$ that they cut out. In particular, these sets are non-empty. Adding more formulae to a partial type corresponds to shrinking the set that they cut out, and vice versa.

However, if we have a *global* type p – a type defined over the entire model $\mathbb{U}$ – then one can no longer identify types with the set $p(\mathbb{U})$ that they cut out, because these sets are usually empty! However, what we can do is *restrict* p to some smaller set A of constants to create a type $p \downharpoonright_A$ over A, defined as the set of all formulae in p that only involve the constants in A. It is easy to see that this is still a type, and if A is small, it cuts out a non-empty set in $\mathbb{U}^n$.

A global type p is said to be *M-invariant*, or *invariant* for short, if the set of formulae in p is invariant under any automorphism σ of $\mathbb{U}$ that fixes M. In particular, given any M-definable set $A \subset \mathbb{U}^n \times \mathbb{U}^m$ and $b \in \mathbb{U}^m$, we see that $p \subset A_b$ if and only if $p \subset A_{\sigma(b)}$, where $A_b := \{a \in \mathbb{U}^n : (a,b) \in A\}$ is a slice of A. (Indeed, this gives an equivalent definition of invariance.)

A trivial example of an invariant global type would be the type of an element $m \in M$ (or in M^n). This cuts out a singleton set $\{m\}$. This is in fact the only invariant global type that cuts out anything at all:

> **Lemma 1** Let p be a global invariant type. Then p is realisable in $\mathbb{U}$ (i.e. $p(\mathbb{U})$ is non-empty) if and only if it is realisable in M (and is the type of a single element in M).

Proof: Suppose p is realisable in $\mathbb{U}$ by some $a \in p(\mathbb{U})$. Since the formula $x = a$ is definable in $\mathbb{U}$, we see that $p \subset \{x : x = a\}$, i.e. p cuts out precisely the singleton set $\{a\}$. As p is invariant, $\{a\}$ must then be invariant under all M-fixing automorphisms of $\mathbb{U}$. By homogeneity, this means that there is no element distinct from a which is elementarily indistinguishable from a over M; in other words, $\{a\}$ is the set cut out by the type $tp(a/M)$ of a over M.

By saturation, the formula $x \neq a$ together with the formulae in $tp(a/M)$ is not satisfiable, hence not finitely satisfiable. Thus there is a finite set of formulae in $tp(a/M)$ that cut out $\{a\}$, i.e. $\{a\}$ is definable over M. But as $\mathbb{U}$ is an elementary extension of M, these formulae must also be realisable in M, i.e. a lies in M, and the claim follows. □

(Because of this, one should regard the notation $p \subset B$ carefully; the set $p(\mathbb{U})$ that p cuts out in the model $\mathbb{U}$ may in fact be empty, but when we write $p \subset B$ for some definable B, we interpret this syntactically rather than semantically (or equivalently, that $p(\mathbb{U}') \subset B(\mathbb{U}')$ holds in all extensions $\mathbb{U}'$ of $\mathbb{U}$, and not just in $\mathbb{U}$ itself.)

On the other hand, invariant global types exist in abundance:

> **Lemma 2** Let p be a type over M. Then there exists an invariant global type q that refines p (i.e. it contains all the formulae that p does).

Proof: We view p as a collection of logically consistent formulae. We enlarge this collection to a larger one p' by adding in the negations of all the formulae definable over $\mathbb{U}$ that are not realisable in M. Observe that this collection remains logically consistent, because any finite set of formulae in p were realisable in $\mathbb{U}$, hence in M (which is an elementary substructure). Hence, by Zorn's lemma, one can extend p' to a global type q.

We now claim that q is invariant. Indeed, let ϕ be a sentence over $\mathbb{U}$ that is contained in q, and let σ be an automorphism that fixes M. If $\sigma(\phi)$ is not in q, then $\neg\sigma(\phi)$ must be in q (by completeness), and hence $\phi \wedge \neg\sigma(\phi)$ is in q also, and hence must be realisable in M (otherwise its negation would be in p', and hence in q). But this is absurd since σ fixes M. Thus $\sigma(\phi)$ does lie in q, yielding invariance.

A major use of invariant global types for us will be that they can be used to generate sequences of indiscernibles (as defined in previous notes):

> **Lemma 3** Let p be a global invariant type of some arity d, and construct recursively a sequence $b_1, b_2, \ldots \in \mathbb{U}^d$ by requiring $b_n \in p \underset{M \cup \{b_1, \ldots, b_{n-1}\}}{\downarrow} (\mathbb{U})$ for all $n = 1, 2, \ldots$. (This is always possible since types are satisfiable once restricted to small sets, by saturation, as discussed earlier). Then $b_1, b_2, \ldots$ are indiscernible over M, i.e. the tuples $(b_{i_1}, \ldots, b_{i_k})$ for $i_1 < \ldots < i_k$ are elementarily indistinguishable (over M) for any fixed k.

Proof: This is achieved by an induction on k. The $k = 1$ case is clear since the b_n all have type $p \underset{M}{\downarrow}$ over M. Now we do the $k = 2$ case. It suffices to show that (b_1, b_2) and (b_i, b_j) are elementarily indistinguishable over M for all $i < j$.

By construction, b_2 and b_j have the same type over $M \cup \{b_1\}$, and so (b_1, b_2) and (b_1, b_j) are elementarily indistinguishable over M. So it remains to show that (b_1, b_j) and (b_i, b_j) are elementarily indistinguishable.

Let A be an M-definable relation that contains (b_1, b_j); we need to show that A contains (b_i, b_j) also.

Since b_1 and b_i have the same type over M, by homogeneity there exists an automorphism σ of $\mathbb{U}$ fixing M that maps b_1 to b_i. Since b_j realises $p \underset{M \cup \{b_1\}}{\downarrow}$, we see that p contains the sentence $(b_1, x) \in A$, hence by invariance p contains $(b_i, x) \in A$ also. Since b_j realises $p \underset{M \cup \{b_i\}}{\downarrow}$, we conclude $(b_i, b_j) \in A$, as required.

This concludes the $k = 2$ case. The higher k case is similar and is left as an exercise. □

— 2. Intersections of wide types —

Now we assume that the structure $\mathbb{U}$ is equipped with an M-invariant Kiesler measure μ. This leads to the notion of a *wide type* – a type such that all the $\mathbb{U}$-definable sets containing this type have positive measure. Intuitively, elements of a wide type are distributed "generically" in the structure.

In the previous notes we showed that wide types can be "split" amongst indiscernables, as follows:

> **Lemma 4** Let b be an element or tuple in $\mathbb{U}$, let a be a wide type over $M \cup \{b\}$ for some set of constants M, and let $b_1, b_2, \ldots$ be a sequence of indiscernibles (over M) that has the same type as b (over M). Then for any finite number $b_1, \ldots, b_n$ in this sequence, one can find a type a' such that a' has the same type over b_i as a does over b, for all $1 \leq i \leq n$.

We now use this lemma to show that sets defined by wide types intersect each other in a uniform fashion.

> **Lemma 5** Let p, q be types over M, and let $a, a' \in p(\mathbb{U})$, $b, b' \in q(\mathbb{U})$ be realisations of a, b such that $tp(a/M \cup \{b\})$ and $tp(a'/M \cup \{b'\})$ are wide. Let A_x, B_y be M-definable sets with parameters, contained inside an M-definable set X of finite measure; then $\mu(A_a \cap B_b) > 0$ if and only if $\mu(A_{a'} \cap B_{b'}) > 0$.

Proof: By homogeneity, there is an automorphism fixing M that sends b to b', and maps a to another element of $p(\mathbb{U})$. Thus without loss of generality we may assume $b = b'$.

We assume for contradiction that $\mu(A_a \cap B_b) > \delta > 0$ and $\mu(A_{a'} \cap B_{b'}) = 0$.

By Lemma 2, we may extend q to an invariant global type q'. Observe that for any $\epsilon > 0$, either one has $\mu(A_a \cap B_x) \geq \epsilon$ for all $x \in q' \downarrow_{M \cup \{a\}}$, or one has $\mu(A_a \cap B_x) \leq \epsilon$ for all $x \in q' \downarrow_{M \cup \{a\}}$ (since there is a $M \cup \{a\}$-definable set between $\{x : \mu(A_a \cap B_x) \geq \epsilon\}$ and $\{x : \mu(A_a \cap B_x) > \epsilon\}$. Suppose first that the former option holds for some ϵ, thus there is a uniform lower bound $\mu(A_a \cap B_x) > \epsilon$. We now define a sequence $a_1, a_2, \ldots \in p(\mathbb{U})$ and an indiscernible sequence $b_1, b_2, \ldots \in q(\mathbb{U})$ as follows:

- We initialise $a_1 = a$ and b_1 to be a realisation of $q' \downarrow_{M \cup \{a_1\}}$.
- Now suppose that $a_1, \ldots, a_n, b_1, \ldots, b_n$ have been chosen with $b_1, \ldots, b_n$ indiscernible. By Lemma 4, we can find $a_{n+1} \in p(\mathbb{U})$ that has the same type over b_i that a' has over b' for all $1 \leq i \leq n$. Since $\mu(A_{a'} \cap B_{b'}) = 0$, this implies that $\mu(A_{a_{n+1}} \cap B_{b_i}) = 0$ for all $1 \leq i \leq n$. (Here we use the fact that $\mu(A_x \cap B_b) = 0$ is a type-definable formula over M and b.)
- Now, let b_{n+1} be a realisation of $q' \downarrow_{M \cup \{a_1, \ldots, a_{n+1}, b_1, \ldots, b_n\}}$. With this construction and Lemma 3 we see by induction that $b_1, \ldots, b_{n+1}$ is also indiscernible; now we iterate the procedure.

Let C_i be the set $C_i := A_{a_i} \cap B_{b_i}$, then observe from the above construction that C_i lies in X and $\mu(C_i \cap C_j) = 0$ for all $i < j$. On the other hand, we claim that $\mu(C_i)$ is uniformly bounded away from zero, this contradicts the finite measure of $\mu(X)$ by the pigeonhole principle.

To see the uniform lower bound, find an automorphism σ_i fixing M that maps a_1 to a_i. By hypothesis, $\mu(A_{a_1} \cap B_{b_1}) > 0$, thus there exists a rational $r > 0$ such that the predicate that models $\mu(A_{a_1} \cap B_x) > r$ is in $\tilde{q}$, hence in q'. By invariance, the predicate $\mu(A_{a_i} \cap B_x) > r$ is in q' also, hence by construction of b_i, $\mu(A_{a_i} \cap B_{b_i}) > r$, and the claim follows.

Now we consider the opposite case, in which $\mu(A_a \cap B_x) = 0$ for all $x \in q' \downarrow_{M \cup \{a\}}$. Then we run the construction slightly differently: for each n in turn, set b_n to be a realisation of $q' \downarrow_{M \cup \{a_1, \ldots, a_{n-1}, b_1, \ldots, b_{n-1}}}$, then set $a_n \in p(\mathbb{U})$ so that $\mu(A_{a_n} \cap B_{b_n}) > \delta$. (This is possible because for any definable set C containing p, the $\bigwedge$-definable set $\{x \in C : \mu(A_x \cap B_b) \geq \delta\}$ contains p and thus has positive measure, and so the same is true for b_n; now use saturation.) Then again we see that the C_i lie in X, have intersection of measure zero, and have measure uniformly bounded from below, and we again obtain a contradiction. ☐

Now we place a group structure on $\mathbb{U}$, and obtain a variant of the above result:

Proposition 6 Let p, q be types in M, with p wide. Suppose that p, q are contained in M-definable sets X, X' such that XX' has finite measure. Let $a, b \in q$ be such that the type of a over $M \cup \{b\}$ is wide. Assume also that the Keisler measure is translation-invariant. Then $pa \cap pb$ is also wide.

Proof: Suppose this is not the case, so that there exists an M-definable set A containing p such that $Aa \cap Ab$ has zero measure. (Initially, one would need two different definable sets containing p, but one can simply take their intersection.) On the other hand, as p is wide, A itself has positive measure. We can place A in X.

By using the fact that wide types over one set of constants can be refined to wide types over larger sets of constants (Lemma 2 from the previous notes), we see that we can recursively construct a sequence $a_1, a_2, ... \in q(\mathbb{U})$ with $tp(a_n/M \cup \{a_1, ..., a_{n-1}\})$ wide for all n. Since $\mu(Aa \cap Ab) = 0$, we conclude from Lemma 5 that $\mu(Aa_n \cap Aa_i) = 0$ for all $1 \leq i < n$. On the other hand, the $\mu(Aa_i)$ all have the same measure as $\mu(A)$, which is positive. Finally, the Aa_i are all contained in XX', which has finite measure; this leads to a contradictoin. ❐

This "generic intersection" property of translates of p will be important in later arguments when creating near-groups.

Comments (4)

 **ayhangunaydin** · 30 October 2009

I think the type p should be invariant in Lemma 1. *[Corrected, thanks – T.]*

 **Terence Tao** · 4 November 2009

Lou van den Dries has kindly made his own notes on the Hrushovski paper available at
http://www.math.uiuc.edu/~vddries/approx.dvi
I understand that they may continue to be updated in the future. (I think that in the next post in this series I will collect all the various links to other notes that people have been supplying.)

 **Link** · 4 November 2009

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